

Topic: Types of Data Structures — Linear and Non-Linear

Q. Classify Data Structures into Linear and Non-Linear types. Differentiate between them and provide examples of each.

 Definition to Remember

A **Data Structure** is a specialized format for organizing, processing, retrieving, and storing data. Based on how elements are arranged, data structures are classified into **Linear** (elements in sequential order) and **Non-Linear** (elements arranged hierarchically or in a network).

1. Linear Data Structures

- Elements are arranged in **sequential order** — each element has a unique predecessor and successor.
- Can be **traversed in a single run**.
- Memory is usually allocated in **contiguous blocks** (arrays) or nodes (linked lists).

Examples:

DS	Description
Array	Contiguous memory block, same data type, index-based access
Linked List	Chain of nodes; each node points to the next
Stack	Follows LIFO (Last In, First Out) principle
Queue	Follows FIFO (First In, First Out) principle

2. Non-Linear Data Structures

- Elements are arranged in a **hierarchical or network** structure.
- One element can be connected to **multiple other elements**.
- Traversal requires **multiple paths or recursive algorithms**.

Examples:

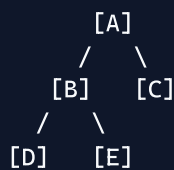
DS	Description
Tree	Hierarchical structure with a root node and child nodes
Graph	Network of vertices (nodes) and edges — no strict hierarchy

3. Linear vs Non-Linear Comparison

Feature	Linear	Non-Linear
Arrangement	Sequential order	Hierarchical / network
Traversal	Single run	Multiple paths / recursion
Memory	Often less efficient	More efficient (dynamic allocation)
Implementation	Simpler	More complex
Levels	Single level	Multi-level
Examples	Array, Stack, Queue, Linked List	Tree, Graph

LINEAR: [A] → [B] → [C] → [D]

NON-LINEAR (Tree):



★ Must-Write Points (for 10 marks)

1. Data structures are classified into **Linear** (sequential) and **Non-Linear** (hierarchical/network).
2. Linear DS: elements arranged sequentially — each has one predecessor and one successor.
3. Linear examples: **Array, Linked List, Stack (LIFO), Queue (FIFO)**.
4. Non-Linear DS: elements connected to multiple others — Tree and Graph.
5. Linear DS can be traversed in a single run; Non-Linear requires multiple paths.
6. Non-Linear DS uses memory more efficiently via dynamic allocation.
7. Trees are used in database indexing; Graphs are used in route planning and AI.

⚡ Quick Recall

Linear (sequential, single run) → Array, Linked List, Stack (LIFO), Queue (FIFO) → Non-Linear (hierarchical/network, multi-path) → Tree, Graph → Non-Linear: efficient memory, complex traversal